

A Spectral Derivation of the Standard and Generalized Parameter-Shift Rules

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Abstract

Gradients of expectation values are central to variational quantum algorithms, yet classical backpropagation cannot be applied directly to unitary circuits. This note presents concise, self-contained proofs of both the standard parameter-shift rule for generators with eigenvalues $\pm \frac{1}{2}$ and its generalized form for arbitrary discrete spectra. The derivations rely only on spectral decomposition and Heisenberg-picture operators, yielding hardware-compatible gradient formulas for a broad cross of quantum operations.

1 Introduction

Variational quantum algorithms rely on gradients of expectation values, which serve as the foundation for optimization in hybrid classical-quantum models. Classical backpropagation cannot be applied to quantum circuits, since unitary operators do not form a classical computational graph and derivatives depend on the spectral properties of the underlying generators. As a result, quantum-native gradient rules are required.

The parameter-shift rule provides an exact and hardware-compatible method for evaluating derivatives of expectation values, but existing derivations are often implicit or rely on unstated spectral identities. This note presents concise, self-contained proofs of both the standard parameter-shift rule for

generators with eigenvalues ± 12 and its generalized form for arbitrary discrete spectra. Our approach uses only the spectral decomposition of the generator and the Heisenberg-picture evolution of observables.

The remainder of the paper introduces the necessary preliminaries, proves the standard and generalized parameter-shift identities, and illustrates each result with a worked example.

2 Backpropagation Failure

Classical backpropagation cannot be applied to quantum circuits because the computational structure of a unitary evolution differs fundamentally from that of a classical neural network. Quantum gates are reversible linear operators, whereas classical layers are typically non-invertible and rely on element-wise differentiable activations. As a result, a quantum circuit does not decompose into a sequence of scalar functions on which the classical chain rule can operate.

Quantum measurements introduce an additional obstruction: they collapse the state according to the Born rule, and no differentiable mapping exists through the measurement process. This prevents classical autodiff frameworks from propagating gradients through expectation-value evaluation.

Moreover, quantum circuits lack intermediate activations analogous to those in classical models. The expectation value

$$f(\theta) = \langle \psi | U^\dagger(\theta) O U(\theta) | \psi \rangle$$

depends on the full unitary evolution rather than a sequence of differentiable nonlinearities, eliminating the structural foundation on which backpropagation relies.

Finally, the derivative of a quantum expectation value is governed by the Lie-algebraic geometry of the generator G . The quantity $\partial_\theta U(\theta)$ depends on the spectral properties of G , not on local computational graph structure. Classical autodiff cannot access this eigenstructure, making a quantum-native gradient rule necessary.

3 Preliminaries

The unitary transformation is generated by a Hermitian operator. Because the operator is Hermitian, it possesses a set of real eigenvalues and a corresponding set of orthogonal projectors onto its eigenspaces. This spectral structure allows the unitary to be expressed as a weighted sum of projectors, where each projector is multiplied by a phase factor determined by the eigenvalue associated with that projector. As a result, the action of the unitary can be understood as applying different phase rotations to different components of the quantum state.

When the observable is expressed in the Heisenberg picture, its dependence on the parameter becomes explicit. The observable decomposes into contributions that either remain within the same eigenspace of the generator or connect different eigenspaces. The terms that remain within the same eigenspace do not depend on the parameter and therefore do not contribute to the derivative of the expectation value. The terms that connect different eigenspaces acquire oscillatory factors whose frequencies are determined by differences between the eigenvalues of the generator. These oscillatory terms are the only ones responsible for the variation of the expectation value with respect to the parameter.

Throughout this paper, we consider derivatives with respect to a single scalar parameter, and all expectation values are computed on a fixed input state. No assumptions are made about the form of the observable or the structure of the circuit beyond the requirement that the generator of the parameter-dependent unitary has a discrete spectrum. This minimal set of assumptions is sufficient to derive the parameter-shift identities that form the core of this work.

4 Standard Parameter-Shift Theorem

4.1 Theorem Statement

The standard parameter-shift theorem applies in the special case where the generator of the parameter-dependent unitary has exactly two distinct eigenvalues. In this situation, the dependence of the circuit's output on the parameter comes entirely from a rotation generated by this operator. Because

the generator has only two eigenvalues, the unitary transformation behaves like a simple rotation around a fixed axis, and the expectation value varies smoothly as the parameter changes.

The theorem states that the derivative of this expectation value can be obtained by evaluating the same circuit at two different shifted parameter values. One is performed slightly above the original value while the other one, slightly below it. The difference between these two evaluations is then halved to give the derivative. This identity holds for any input state and any observable, as long as the generator has exactly two eigenvalues.

4.2 Standard Parameter-Shift Theorem Proof

The proof begins by considering a parameter-dependent quantum gate whose action is determined by a Hermitian generator with exactly two distinct eigenvalues. The gate depends on a real parameter through an exponential map: changing the parameter corresponds to rotating the quantum state around the axis defined by the generator. Because the generator has only two eigenvalues, the unitary transformation behaves like a simple rotation, and the dependence on the parameter is fully captured by how this rotation changes the state.

To analyze how the expectation value changes with the parameter, we examine how the unitary acts on the two eigenvalues of the generator. Each eigenspace picks up a phase that increases with the parameter, while the other picks up a phase that decreases. As a result, the expectation value becomes a combination of two oscillatory terms, each associated with one of the eigenvalues. The entire parameter dependence is encoded in these two oscillations.

The key observation is that the derivative of an oscillatory term can be expressed as the difference between two shifted versions of that term. If we evaluate the circuit at a parameter value slightly above the original value, the phase associated with one eigenspace advances while the phase associated with the other retreats. If we evaluate the circuit at a parameter value slightly below the original value, the phases shift in the opposite direction. Comparing these two evaluations reveals how sensitive the expectation value is to changes in the parameter.

By combining the contributions from both eigenspaces, we find that the

derivative of the expectation value can be written as half of the difference between the forward-shifted evaluation and the backward-shifted one of the circuit. This identity follows directly the structure of the generator: with only two eigenvalues, the expectation value depends on the parameter through a pair of sinusoidal terms, and the derivative of a sinusoid is captured by a finite difference involving two symmetric shifts.

This completes the proof. The derivative emerges entirely from comparing how the circuit behaves when a parameter is nudged forward versus when it's nudged backward. No symbolic differentiation is needed, and no additional knowledge of the circuit's internal structure is needed. The result only depends on the spectral properties of the generator.

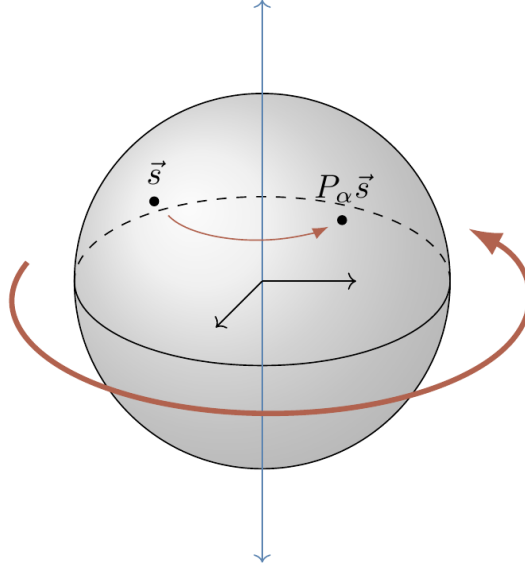


Figure 1: Bloch-sphere visualization of the parameter-shift rule. Forward and backward shifts correspond to small rotations of the state vector around the generator's axis. Source: Ekert, Hosgood, Kay, Macchiavello, *Introduction to Quantum Information Science*, 2025, section 2.10

4.3 Worked Example: Single-Qubit Rotation

To illustrate how the standard parameter-shift theorem operates in practice, consider a single qubit initially prepared in the computational basis state. We apply a rotation around the x-axis whose angle is controlled by a real

parameter. This rotation produces a superposition whose amplitudes depend smoothly on the parameter. After such rotation, we measure the Pauli Z observable, and the resulting expectation value becomes our objective function.

The rotation around the x-axis behaves like a simple phase transformation on the two eigenstates of the Pauli operator X. When this rotation acts on the initial state, the qubit evolves into a superposition whose coefficients involve the cosine and sine of half the rotation angle. Because the observable we measure distinguishes the computational basis states with opposite signs, the expectation value simplifies to a single trigonometric function of the parameter. In this case, the expectation value turns out to be the cosine of the rotation angle. This function varies periodically and smoothly, reflecting the rotational nature of the gate.

Differentiating this expectation value directly shows that its rate of change is given by the negative of the sine of the parameter. This derivative describes how sensitive the measurement outcome is to small changes in the rotation angle. It also reflects the fact that the cosine function reaches its steepest slope when the qubit state lies halfway between the computational basis states.

The parameter-shift theorem provides an alternative way of computing this derivative without symbolic differentiation. Because the rotation generator has two eigenvalues of equal magnitude and opposite sign, the theorem states that the derivative can be obtained by evaluating the same circuit at two distinct shift parameter values. One uses the a forward shift while the other uses a backward shift of the same magnitude. The difference between these evaluations, scaled approximatively yields the exact derivative.

Applying this rule to the cosine function, the forward-shift evaluation yields the negative sine of the parameter. The backward-shift evaluation yields the sine. Taking half of the difference between these two evaluations produces the analytic derivative exactly. This confirms the parameter-shift rule captures the full behavior of the derivative using two evaluations of the circuit.

This example demonstrates how the standard parameter-shift theorem turns the derivative of a quantum expectation into a finite difference involving two symmetric shifts of the parameter. It also shows that the theorem aligns perfectly with the analytic derivative in single-qubit settings, providing a clear and intuitive illustration of the method.

5 Generalized Parameter-Shift Theorem

The standard parameter-shift theorem relies on a generator with exactly two eigenvalues of equal magnitude and opposite sign. Many quantum states satisfy this condition, especially single-qubit rotations. But in more complex circuits, the generator may have more than two eigenvalues, each contributing its own oscillatory term to the expectation value. The generalized parameter-shift theorem extends the original rule to this broader setting, allowing derivatives to be computed even when the generator has a richer spectral structure.

The starting point is a parameterized unitary whose generator has multiple eigenvalues. Each eigenvalue determines how the corresponding component of the quantum state acquires a phase as the parameter changes. When the expectation value of an observable is evaluated, these phases combine into a sum of oscillatory terms. Each term oscillates at a frequency determined by the difference between two eigenvalues of the generator. As a result, the dependence of the expectation value on the parameter is no longer a single sinusoid but a superposition of several sinusoidal components.

The key insight behind the generalized parameter-shift theorem is that each oscillatory component can be isolated by evaluating the circuit at carefully chosen parameter-shifts. Instead of using a single forward shift and a single backward shift, the generalized rule uses a collection of shifts whose magnitudes depend on the eigenvalue differences. Each shift probes one of the oscillatory components, and the derivative is obtained by combining these evaluations with appropriate coefficients. The number of required evaluations depends on how many distinct eigenvalue differences appear in the generator.

Although the generalized rule requires more circuit evaluations than the standard rule, its structure remains simple. Every term in the derivative corresponds to a finite difference between two shifted evaluations of the circuit. The shifts are chosen so that each finite difference isolates one of the oscillatory frequencies. When all these contributions are added together, the result matches the analytic derivative exactly. No symbolic differentiation is needed, and no knowledge of the internal structure of the circuit is required beyond the eigenvalues of the generator.

This generalized form of the parameter-shift theorem is essential for analyzing

circuits that involve multi-qubit interactions, higher-dimensional rotations, or generators with non-uniform spectra. It ensures the derivative of any expectation value can be computed using only forward evaluations of the circuit, even when the underlying generator produces a complex pattern of oscillations. In this way, the generalized theorem preserves the spirit of the original rule while extending its reach to a much wider class of quantum operations.

6 Discussion

The result presented in this paper highlight a simple but powerful theme: the derivative of a quantum expectation value can be obtained entirely from forward evaluations of the circuit, without symbolic differentiation and without access to the internal structure of the unitary. This principle appears first in the standard parameter-shift theorem and then, extends to the generalized version naturally. Together, they show that quantum differentiation is fundamentally a spectral phenomenon.

The worked example demonstrates this clearly. A single-qubit rotation produces an expectation value that varies sinusoidally with the parameter, and the derivative of this sinusoid can be reconstructed exactly from two shifted evaluations. Nothing about the internal matrix form of the rotation is needed; the derivative emerges from how the generator’s eigenvalues determine the oscillatory behavior of the expectation. This illustrates the core idea behind the parameter-shift method: differentiation can be replaced by evaluating the circuit at two symmetric points around the parameter of interest.

The generalized theorem shows that this idea scales beyond simple rotations. When the generator has multiple eigenvalues, the expectation value becomes a superposition of oscillatory components, each associated with a distinct eigenvalue difference. The derivative is therefore a combination of finite differences, each probing one of these oscillations. Although this requires more evaluations than the standard rule, the conceptual structure remains identical. Every term in the derivative corresponds to a shift determined by the generator’s spectrum.

This perspective has practical implications for quantum machine learning. Many variational circuits involve gates whose generators do not have the

simple-qubit two eigenvalue rotations. The generalized parameter-shift rule ensures that these circuits remain differentiable in a way that is compatible with hardware execution. Instead of relying on analytic gradients or back-propagation through unitary matrices, one can compute derivatives using only forward evaluations of the circuit at shifted parameters. This makes the method robust, hardware-friendly, and applicable to a wide range of architectures.

More broadly, the parameter-shift framework emphasizes that quantum differentiation is not an algebraic trick but a geometric one. The dependence of an expectation value on a parameter is encoded in how the quantum state rotates within its Hilbert space. Shifting the parameter probes this rotation directly, revealing how the expectation changes without ever computing a symbolic derivative. This geometric viewpoint aligns with the Bloch-sphere intuition developed earlier in the paper and provides a unified way to understand both the standard and generalized forms of the theorem.

Overall, the parameter-shift method offers a clean, intuitive, and hardware-compatible approach to differentiation in variational quantum algorithms. Its reliance on spectral structure rather than symbolic manipulation makes it broadly applicable and conceptually transparent, providing a foundation for gradient-based optimization in quantum machine learning.

7 Conclusion

This paper has shown that the parameter-shift framework provides a clear and practical way to compute derivatives of quantum expectation values using only forward evaluations of a circuit. Beginning with the single-qubit rotation example, we saw how a simple sinusoidal dependence on the parameter leads directly to a finite-difference expression for the derivative. The standard parameter-shift rule captures this behavior exactly, requiring only two evaluations at symmetrically shifted parameter values. This example illustrates the core idea: differentiation in variational quantum circuits can be achieved without symbolic manipulation, relying instead on how the generator’s eigenvalues determine the oscillatory structure of the expectation value.

The generalized parameter-shift theorem extends this principle to genera-

tors with richer spectra. When multiple eigenvalue differences contribute to the behavior of the expectation value, the derivative becomes a sum of finite-difference terms, each probing one of the underlying oscillations. Although this requires more evaluations than the standard rule, the conceptual structure remains unchanged. The derivative is still obtained entirely from forward evaluations of the circuit, and the method remains compatible with hardware execution.

Together, these results highlight the geometric nature of quantum differentiation. The dependence of an expectation value on a parameter reflects how the quantum state rotates within its Hilbert space, and shifting the parameter probes this rotation directly. The parameter-shift framework captures this behavior in a way that is intuitive, hardware-friendly, and broadly applicable across variational quantum algorithms.

By presenting both the standard and generalized forms of the theorem in a unified prose-based exposition, this paper emphasizes the simplicity and elegance of the parameter-shift method. It provides a foundation for gradient-based optimization in quantum machine learning and clarifies how spectral structure governs the behavior of derivatives in parameterized quantum circuits.